

POWERMATIC 1200 RETROFIT - CLICK PLUS LADDER LOGIC

Programming reference generated from Powermatic_1200_Retrofit_Documentation_rJ.md section 7.
 Key Rev J jog rules: any STOP press exits jog; drum OFF inhibits motion but not jog mode;
 the 5 s FWD entry hold is consumed by C50 and cannot command spindle motion.

7. Ladder logic - CLICK PLUS (see DWG 05)

Transcribe into the free CLICK Programming Software. Symbols:

-] [- N.O. · -]/[- N.C. · -]UP[- leading-edge (press) · -]DN[- trailing-edge (release) ·
 -()- OUT · -(S)- SET · -(R)- RST · [TMR ...] timer.

> Scan order is load-bearing in Rev J. The bidirectional stop-first behavior and the jog-mode
 > short-press logic depend on rungs executing in the order listed. Two rules to preserve: (a)
 the DRILL
 > latch-SET rungs (4, 5) run before the opposite-stop rung (7); (b) the jog/consume flag C50
 > is set by the 5 s entry hold before the jog output rung can act on that held FWD input.
 Transcribe in
 > the numbered order and don't reorder.

RUNG 1 · RUN PERMISSIVE -> C20 (KA proved · no immediate trip · drum ON · STOP clear ·
 healthy)

```

X011      X013      X008      X004      C10      C11
--] [-----]/[-----+--] [-----+--] [-----]/[-----]/[----- ( )-- C20
          | X009 |
          +--] [-----+

```

C20 = X011 AND NOT X013 AND (X008 OR X009) AND X004 AND NOT C10 AND NOT C11

RUNG 2 · SPEED PRESET -> Y003 (drum LOW -> 30 Hz; jog forces LOW)

```

X008
--] [---+----- ( )-- Y003 PRESET_LOW
C60 | (jog armed -> force 30 Hz)
--] [---+

```

RUNG 3 · BLINK ASTABLE -> C99 (~1 Hz; or substitute a CLICK built-in clock relay)

```

C99
--]/[-----[TMR Tb1 K500ms]
Tb1
--] [----- (S)-- C99
C99
--] [-----[TMR Tb2 K500ms]
Tb2
--] [----- (R)-- C99

```

----- DRILL - latched, bidirectional

RUNG 4 · DRILL FWD - LATCH SET on RELEASE (short tap latches; NOT C50 = not consumed by a
 long-hold/opposite-stop; NOT C60 = not in jog)

```

X001      X003      C50      C60      C20      C10      C31
--] [-----]DN[-----]/[-----]/[-----]/[-----]/[----- (S)-- C30 DRILL_RUN
      release NOT consumed NOT jog permit noFault NOT REV

```

RUNG 5 · DRILL REV - LATCH SET on PRESS (mirror of FWD; no long-hold role, so acts on
 press)

```

X001      X005      C60      C20      C10      C30
--] [-----]UP[-----]/[-----]/[-----]/[-----]/[----- (S)-- C31 DRILL_REV
      press NOT jog permit noFault NOT FWD

```

RUNG 6 · DRILL - LATCH RESET (stop / not DRILL / lost permit / fault -> drop both)

```

X004
--]/[---+-----+----- (R)-- C30
X001 | +----- (R)-- C31
--]/[---+ (NOT DRILL)
C20 | (NOT permit - incl. X011 safety, drum-OFF, over-temp)
--]/[---+
C10 | (fault)
--] [---+

```

RUNG 7 · OPPOSITE-AS-STOP (opposite button = stop only; runs AFTER the SET rungs)

```

X005      C30
--]UP[-----] [-----+----- (R)-- C30 REV press stops a FWD run
          +----- (S)-- C50 ...and consume, so its release can't
          latch REV path

X003      C31
--]UP[-----] [-----+----- (R)-- C31 FWD press stops a REV run
          +----- (S)-- C50 ...and consume, so its RELEASE can't
          latch FWD

```

(Why it works: RUNG 5's NOT C30 / RUNG 4's NOT C31 interlocks block the opposite latch on the
 same press
 because the running latch is still set when SET evaluates; RUNG 7 then drops it. Second
 press latches.)

RUNG 8 · CLEAR FWD-CONSUMED ON RELEASE

```

X003
--]/[----- (R)-- C50

```

```

RUNG 9 · DRILL BOTTOM-LIMIT FAULT    (FWD only; reverse is not a feed-down runaway)
  X001      C30      X006
--] [-----] [-----]/[-----](S)-- C10  FAULT

----- TAP - single-tap state machine -----
-----

RUNG 10 · TAP START (IDLE -> TAP_DOWN)  entry gate: lever OFF (X010=1), not in jog
  X002      X003      X010      C60      C20      C10      C2      C3
--] [----]UP[----] [----]/[----] [----]/[----]/[----]/[----](S)-- C2  TAP_DOWN
                        NOT jog      permit  noFault

RUNG 11 · TAP DOWN -> BACK_OUT    (bottom limit)
  C2      X006
--] [-----]/[-----+------(R)-- C2
                        +------(S)-- C3  BACK_OUT

RUNG 12 · TAP BACK_OUT -> IDLE    (top limit)      [note C: SET C2 here for continuous]
  C3      X007
--] [-----]/[------(R)-- C3

RUNG 13 · TAP ABORT    (stop / not TAP / fault / lost-permit-or-safety) ← Rev H: adds NOT C20
  leg
  X004
--]/[---+-----+------(R)-- C2
  X002 |                      +------(R)-- C3
--]/[---+ (NOT TAP)
  C10 | (fault)
--] [---+
  C20 | (lost permit - incl. E-stop/safety via X011, drum-OFF, STOP)
--]/[---+

RUNG 14 · TAP FEED-LEVER RUNTIME TRIP    (lever ON during a cycle = tap-snap)
  C2      X010
--] [---+--]/[------(S)-- C10  FAULT
  C3      | LeverON
--] [---+

----- DB over-temp - self-clearing -----

RUNG 15 · DB OVER-TEMP SET    (X012=0 = over-temp OR wire break -> assert immediately)
  X012
--]/[------(S)-- C11  THERMAL

RUNG 16 · DB COOL-DOWN -> SELF-CLEAR    (clears only after X012 stays normal for the cool-down
  delay)
  X012
--] [-----[TMR Tcool K10000ms]
  Tcool
--] [------(R)-- C11
  (Self-clearing: no manual reset, and C11 is NOT the operator-clearable C10 - so it can't
  fight RUNG 17.)

RUNG 17 · FAULT RESET    (STOP while fully stopped clears the hard-fault latch)
  X004      C2      C3      C30      C31
--]/[-----]/[-----]/[-----]/[------(R)-- C10  FAULT

----- JOG mode -----

RUNG 18 · FWD HOLD TIMER    (times how long FWD is held)
  X003
--] [-----[TMR T5FWD K5000ms]

RUNG 19 · ENTER JOG    (FWD held >=5 s, safe, not already armed; consume the press so release
  won't latch a run)
  T5FWD      C50      C60      X011      X013      C10
--] [----]/[-----]/[-----] [----]/[-----]/[-----+-(S)-- C60  JOG_MODE
                        +-(S)-- C50  (consumed)

RUNG 20 · JOG IDLE TIMEOUT    (60 s with no jog button -> auto-exit)
  C60      X003      X005
--] [----]/[-----]/[-----[TMR T60 K60000ms]
  T60
--] [------(R)-- C60

RUNG 21 · JOG FORCE-EXIT    (any latched fault or safety trip)
  C10
--] [---+------(R)-- C60
  X011 |
--]/[---+ (final safe state lost)
  X013 |
--] [---+ (immediate safety trip)

RUNG 22 · STOP EXIT (jog)    (any STOP press exits jog mode)
  C60      X004
--] [-----]/[------(R)-- C60

```

RUNG 23 · RESERVED (Rev J deletes the former STOP-hold distinction)

```
RUNG 23A · JOG RUN PERMISSIVE -> C61
  C60      X011      X013      X008      X004      C10      C11
--] [----] [-----]/[-----+--] [---+---] [-----]/[-----]/[-----] ( )-- C61
                        |   X009   |
                        +---] [----+
```

----- VFD run outputs -----

```
RUNG 24 · VFD RUN-FWD -> Y001
  C30      C20      Y002
--] [---+---] [-----]/[-----] ( )-- Y001 VFD_FWD
  C2      | (normal: DRILL_RUN or TAP_DOWN)
--] [---+
  C61      | X003      C50      Y002
--] [---+--] [-----]/[-----]/[--- (jog-FWD: NOT C50 means the 5 s entry hold can NOT jog)

Y001 = [ (C30 OR C2) AND C20 AND NOT Y002 ] OR [ C61 AND X003 AND NOT C50 AND NOT Y002 ]
```

```
RUNG 25 · VFD RUN-REV -> Y002
  C31      C20      Y001
--] [---+---] [-----]/[-----] ( )-- Y002 VFD_REV
  C3      | (normal: DRILL_REV or TAP back-out)
--] [---+
  C61      | X005      Y001
--] [---+--] [-----]/[--- (jog-REV)

Y002 = [ (C31 OR C3) AND C20 AND NOT Y001 ] OR [ C61 AND X005 AND NOT Y001 ]
```

----- Chunk (Contactor B, theater) -----

```
RUNG 26 · CHUNK REQUEST (edge of entering any run / direction - TAP or DRILL)
  C2
--]UP[---+------(S)-- C40 CHUNK_REQ
  C3      |
--]UP[---+
  C30      |
--]UP[---+
  C31      |
--]UP[---+
```

```
RUNG 27 · CHUNK TIMER + OUTPUT
  C40
--] [-----[TMR T1 K300ms]
  C40      T1
--] [-----]/[-----] ( )-- Y004 CONTACTOR_B
  T1
--] [------(R)-- C40
```

----- Indicators (4 states, exactly one active) -----

```
RUNG 28 · GREEN -> Y005 (solid = all-good; blink = jog armed)
  X011      X013      C10      C11      C60
--] [-----]/[-----]/[-----]/[-----] ( )-- Y005 GREEN
  X011      | X013      C10      C11      C60      C99
--] [---+--]/[-----]/[-----]/[-----] [--- (jog: blink via C99)
```

```
RUNG 29 · RED -> Y006 (solid = safety/hard fault; blink = over-temp)
  X011
--]/[---+------( )-- Y006 RED
  X013      |
--] [---+ (immediate safety trip)
  C10      | (NOT final-safe OR immediate-trip OR fault -> solid)
--] [---+
  C11      | X011      X013      C10      C99
--] [---+--] [-----]/[-----]/[-----] [--- (over-temp blink only when no solid-red)
```

Ladder notes

- Note A - force LOW in TAP (optional). Rung 2 already Y003 = X008 OR C60 for jog. To also force LOW in every TAP cycle, add OR C2 OR C3. Left off so the drum is honored in TAP.
- Note B - FWD/REV interlock. Rungs 24/25 each carry the other output's N.C.; the drive ramps through zero on reversal, so no PLC dead-time is required. Direction *changes* in DRILL are stop-first (RUNGs 5/6/7), so the interlock only ever guards simultaneous-edge corner cases.
- Note C - continuous tapping. In Rung 12 also SET C2 when it RST C3 (top -> back down). One rung.
- Note D - first scan. IDLE is derived and every latch (C2 C3 C10 C11 C30 C31 C60) comes up cleared, so the machine powers up stopped, out of jog, and valid - no init rung.
- Note E - max-dwell watchdog (optional). Time C2/C3; if a state overruns its limit, SET C10. Backstops the VFD over-torque trip and E-stop; does not replace them.
- Note F - C50 is a shared "this FWD press is spent" flag. It is set either when a 5 s hold enters jog

(RUNG 19) or when a FWD press is used to stop a running REV (RUNG 7). Either way the FWD release in RUNG 4 sees C50=1 and does not latch a forward run. RUNG 24 also requires NOT C50, so the 5 s entry hold cannot start the spindle while the button is still held. C50 clears on FWD release (RUNG 8), after which a fresh FWD or REV press can command jog motion.

- Note G - blink source. RUNG 3 is a portable 1 Hz astable. If you prefer, CLICK PLUS exposes a built-in clock-pulse relay you can use for C99 instead and delete RUNG 3.
- Note H - jog is advisory. Nothing in the jog rungs initiates the safety stop. An E-stop immediately opens the hardwired FWD/REV paths; after the validated delay, KA and both STO channels open and X011 falls. RUNG 21 then force-exits jog on NOT X011.
- Note I - jog exits on STOP. STOP is not overloaded with a long-hold distinction in jog mode. Any STOP press resets C60, stops motion, and returns the pendant to normal DRILL/TAP behavior.
